FOUNDATIONS

Classical ciphers

Cryptography lecture series

An introduction to cryptography

Historical constructions, mathematical definitions, and worked cryptanalysis.

LECTURE 01

Master’s level • Mathematics

Three hours, including two 10-minute breaks • Revised edition

FOUNDATIONS

Why would we want secret writing?

Alice wants to send Bob a message. Eve may read whatever crosses the communication channel.

THE MESSAGE

```
Meet at the north gate at noon.
```

Can Alice make the intercepted message unintelligible to Eve while allowing Bob to recover it?

Start with the communication problem; the construction comes next.

What is cryptography?

Cryptography studies mathematical methods for protecting information in the presence of adversaries.

Cryptanalysis studies how to defeat a proposed protection under specified assumptions.

Today we focus on **confidentiality**: preventing an interceptor from learning the message.

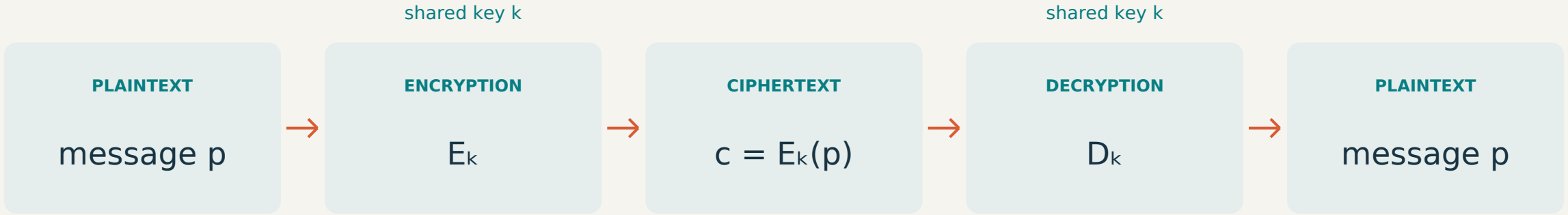
Other goals, including authenticity and integrity, belong to later lectures.

Plaintext, ciphertext, algorithm, key

Term	Meaning in today’s setting
Plaintext	The message before encryption.
Ciphertext	The transformed message sent over the channel.
Cipher / algorithm	The rule used to encrypt and decrypt.
Key	The parameter that selects a particular transformation.

Knowing a ciphertext is different from knowing the key that produced it.

Encryption and decryption



Alice encrypts using a key. Bob uses the corresponding key to undo the transformation.

Eve observes the ciphertext; she should not be able to recover the message without the required secret.

Symmetric-key encryption

Alice and Bob share a secret key before they communicate.

All five ciphers in this lecture are **symmetric-key** constructions: both operations are determined by that shared key.

Distributing and protecting the key is a separate task. For now, assume Alice and Bob already share it.

The algorithm can be public; the key is the secret parameter.

A mathematical model of a cipher

Specify a plaintext space, a ciphertext space, and a key space.

$$\mathcal{P}, \quad \mathcal{C}, \quad \mathcal{K}; \qquad E_k : \mathcal{P} \to \mathcal{C}, \quad D_k : \mathcal{C} \to \mathcal{P}$$

Correctness means that decryption recovers every allowed plaintext:

$$D_k(E_k(p)) = p \qquad \text{for every } p \in \mathcal{P} \text{ and } k \in \mathcal{K}.$$

We will identify these objects explicitly for each construction.

What does the attacker know?

Assume Eve knows the cipher algorithm, but not the particular key.

Information available	Attack setting
Only intercepted ciphertext	Ciphertext-only
Some matching plaintext and ciphertext	Known-plaintext
Encryptions of messages selected by Eve	Chosen-plaintext

Our main substitution and Vigenère attacks use ciphertext-only information.

Public-algorithm principle: Kerckhoffs, La Cryptographie Militaire (1883).

The alphabet as a mathematical object

LETTER ENCODING

A	B	C	D	E	…	X	Y	Z
0	1	2	3	4	…	23	24	25

$$\mathbb{Z}_{26} = \{0, 1, \ldots, 25\}, \qquad \mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}$$

Normalize examples to letters only. We use uppercase for message rows; partial substitution solutions use lowercase.

Removing spaces or merging letters is preprocessing, and may discard information.

Modulo 26: the alphabet wraps around

$$(24 + 2) \bmod 26 = 0, \qquad (0 - 2) \bmod 26 = 24$$

LETTER INTERPRETATION

`Y + 2 → A` `A − 2 → Y`

We represent every residue by an integer from 0 to 25.

When using indexed messages, positions start at 0. Permutation tables later explicitly use positions 1 through m.

Historical motivation and our route

Construction	Historical / mathematical idea
Shift	A small offset makes hand encryption simple.
Substitution	An arbitrary alphabet gives many more keys.
Vigenère	Change the alphabet with the key position.
Playfair	Encrypt pairs using a compact square.
Transposition	Rearrange positions instead of changing letters.

We will understand each construction before examining its limitations.

Plan for the three hours

Elapsed time	Topic
00–30 min	Foundations, shift, and substitution
30–60 min	A complete guided frequency-analysis example
60–70 min	Break
70–120 min	Vigenère, Kasiski, IC, and key recovery
120–130 min	Break
130–172 min	Playfair and transposition
172–180 min	Comparison, reflection, and reading

Short exercises are included in these allocations. Appendix tables support the worked examples.

SHIFT

01 / Shift cipher

Use one fixed offset throughout the message.

A familiar historical example is Caesar’s shift by three positions.

The offset is the key; the alphabet and rule are public.

Shift cipher: definition

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = \mathbb{Z}_{26}$$

$$c_i = E_k(p)_i = (p_i + k) \bmod 26$$

$$p_i = D_k(c)_i = (c_i - k) \bmod 26$$

There are 26 mathematical keys. The key 0 is the identity, leaving 25 nontrivial shifts.

Why does decryption work?

Encryption adds the same residue that decryption subtracts.

$$D_k(E_k(p))_i = ((p_i + k) - k) \bmod 26 = p_i$$

WRAP-AROUND CHECK

```
Y → A with k = 2
A → Y when we subtract 2
```

The equality is in the residue class ring; our displayed values are its representatives.

Shift example: compute the first letters

PLAINTEXT AND KEY

CRYPTOGRAPHYISFUN k = 2

Plaintext	Numeric calculation	Ciphertext
C = 2	(2 + 2) mod 26 = 4	E
R = 17	(17 + 2) mod 26 = 19	T
Y = 24	(24 + 2) mod 26 = 0	A
P = 15	(15 + 2) mod 26 = 17	R

Compute before revealing the complete row on the next slide.

Shift example: finish and reverse

ALIGNED MESSAGE ROWS

```
Plaintext   CRYPTOGRAPHYISFUN
Ciphertext  ETARVQITCRJAKUHWP
```

$$(4-2) \bmod 26 = 2\ (C), \qquad (0-2) \bmod 26 = 24\ (Y)$$

Apply the same subtraction to every ciphertext symbol to recover the normalized plaintext.

Recovering a shift key

Candidate k	Decryption of KHOORZRUOG
1	JGNNQYQTNF
2	IFMMPXPSME
3	HELLOWORLD
4	GDKKNVNQKC

Exhaustive search has only 26 candidates. Readable language helps choose among them.

$$\text{Known } p_i, c_i \quad \Longrightarrow \quad k = (c_i - p_i) \bmod 26.$$

Exercise: recover the message

Individual calculation • 2 minutes

INTERCEPTED

```
PHHWPHDIWHUFODVV
```

Find a plausible key and plaintext. Explain the evidence for your choice.

If you are told M encrypts to P, recover the key directly.

An answer and calculation are in the instructor guide.

02 / Substitution cipher

Replace the fixed shift by an arbitrary permutation of the alphabet.

Every occurrence of one plaintext letter still receives the same ciphertext letter.

Historical appeal: a much larger family of secret alphabets.

Substitution: a bijection and its inverse

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = S_{26}$$

Here S_{26} is the set of all permutations of the 26 alphabet symbols.

$$c_i = \sigma(p_i), \qquad p_i = \sigma^{-1}(c_i)$$

$$\sigma^{-1}(\sigma(p_i)) = p_i.$$

A one-to-one map is essential: a collision would prevent unique decryption.

Substitution example: use the table

INPUT ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

OUTPUT ALPHABET

QWERTYUIOPASDFGHJKLZXCVBNM

Input letters	Look up each image	Result
B E C A U S E	W T E Q X L T	WTEQXLT

Decrypt W by finding it in the output row: it lies under B.

Counting keys and preserving patterns

$$|\mathcal{K}| = 26 \cdot 25 \cdots 2 \cdot 1 = 26! \approx 4.03 \times 10^{26} \approx 2^{88.4}$$

Trying the entire space at one billion keys per second takes approximately 12.8 billion years.

$$p_i = p_j \quad \Longleftrightarrow \quad \sigma(p_i) = \sigma(p_j)$$

A PATTERN THAT SURVIVES

LETTER → STZZTK

A large key space rules out one simple attack; it does not rule out every attack.

Frequency analysis: why counting helps

A substitution changes the labels of the letters, but not their occurrence counts.

$$f_C(\sigma(a)) = f_P(a), \qquad \frac{f_C(\sigma(a))}{N} = \frac{f_P(a)}{N}$$

If the language has characteristic frequencies, the ciphertext inherits them under unknown labels.

Historical context: al-Kindi described frequency-based cryptanalysis in the 9th century.

A language model is a guide

Letter	E	T	A	O	I	N
Illustrative frequency	12.7%	9.1%	8.2%	7.5%	7.0%	6.7%

These are approximate English frequencies, not exact properties of every message.

Pairs such as TH, HE, IN, ER and triples such as THE, AND, ING provide additional evidence.

Absence of a common pair in a short message is weak evidence, not a logical impossibility.

Our intercepted message

168 LETTERS • ORIGINAL REFERENCE EXAMPLE

```
YIFQFMZRWQFYVECFMDZPCVMRZWNMDZVEJBTXCDDUMJ
NDIFEFMDZCDMQZKCEYFCJMYRNCWJCSZREXCHZUNMXZ
NZUCDRJXYYSMRTMEYIFZWDYVZVYFZUMRZCRWNZDZJJ
XZWGCHSMRNMDHNCMFQCHZJMXJZWIEJYUCFWDJNZDIR
```

Assume English plaintext. The unknown substitution key differs from the earlier illustrative table.

We will recover the message incrementally, keeping a partial key and checking each hypothesis.

Ciphertext and plaintext retained from the supplied deck, slides 18–36.

Step 1: count the letters

Ciphertext letter	Z	M	C	D	F	J
Count	20	16	15	13	11	11
Frequency	0.119	0.095	0.089	0.077	0.065	0.065

$$\widehat{p}_M = \frac{16}{168} \approx 0.0952, \qquad \widehat{p}_Z = \frac{20}{168} \approx 0.1190$$

Z is a reasonable candidate for plaintext e. M could correspond to several common letters.

This proposes Z → e; it does not establish that mapping with certainty.

Step 2: try Z → e

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····e··········e····e···e··········
·······e····e·················e····e···e
·e·················e···e···e···e···e·e··
·e·····················e···e··········e···
```

Write an e at every occurrence of Z. Leave every other position undetermined.

Now examine pairs involving Z: if Z means e, then ZW means e? and WZ means ?e.

Maintain consistency by applying a proposed mapping to the whole message.

Step 3: use ZW and WZ

Observed pair	Count	If Z → e
ZW	4	e?
WZ	0	?e
W alone	8	8/168 ≈ 0.0476

Common continuations of e include r, s, n, d, a, t. The asymmetry and W’s frequency make d a useful candidate to test.

Adopt W → d provisionally. A zero WZ count does not prove that alternatives are impossible.

We combine imperfect clues rather than treating one statistic as decisive.

Step 4: test R → n

RW occurs twice. With W → d, it becomes ?d.

The common pairs nd and ed suggest n or e. If Z → e is retained, injectivity prevents R → e.

PARTIAL KEY

Z → e W → d R → n

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····end·········e····ned···e···········
·······e····e·········n··d···en····e····e
·e···n·····n·····ed···e···e··ne·nd·e·e··
·ed····n···········e····ed·····d···e··n
```

The exclusion uses bijectivity; the initial language guess remains provisional.

Step 5: test N → h

Pair	Actual count	Candidate reading
NZ	3	he if N → h
ZN	1	eh if N → h

HE is common in English. Try N → h and inspect the emerging fragments.

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····end·········e····nedh··e···········
h·····e····e·········nh·d···en····e·h··e
he···n·····n·····ed···e···e··ne·ndhe·e··
·ed····nh···h·····e····ed·····d··he··n
```

Counts are recomputed from the displayed ciphertext, including the one occurrence of ZN.

Step 6: recognize a candidate AND

BEFORE AND AFTER THE HYPOTHESIS

```
RZCRWNZ → ne·ndhe
RZCRWNZ → neandhe
```

The middle pattern ·nd suggests and. This proposes C → a.

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····end····a···e·a··nedh··e·····a·····
h·····ea···e·a···a···nhad·a·en··a·e·h··e
he·a·n·····n·····ed···e···e··neandhe·e··
·ed·a···nh···ha···a·e····ed·····a·d··he··n
```

A plausible word boundary gives “…ne and he…”, but must agree with the rest of the text.

Step 7: choose a candidate for M

M occurs 16 times. We will test the candidate M → i.

RNM becomes nh·, suggesting a word boundary before h followed by a vowel. CM becomes a·; ai is more plausible than ao.

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····iend·····a·i·e·a·inedhi·e·····a···i·
h·····i·ea·i·e·a···a·i·nhad·a·en··a·e·hi·e
he·a·n·····in·i····ed···e···e·ineandhe·e··
·ed·a··inhi··hai··a·e·i··ed·····a·d··he··n
```

These language arguments rank hypotheses; they do not constitute a proof.

Step 8: recover a likely t

JN occurs twice. Since N → h, the pair becomes ·h.

TH is a common English pair. Try J → t and apply it everywhere.

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····iend·····a·i·e·a·inedhi·e··t···a···it
h·····i·ea·i·e·a···ati·nhadta·en··a·e·hi·e
he·a·nt····in·i····ed···e···e·ineandhe·ett
·ed·a··inhi··hai··a·eti·ted··t··a·d·the··n
```

Emerging fragments include “…had ta·en…” and “…and he ·ett·ed…”.

Step 9: complete some word patterns

The fragment ·oo·in· suggests looking: Y → o is a useful test. Four occurrences of MD become i·; D → s gives is.

PARTIAL PLAINTEXT • · = UNDETERMINED

```
o····iend··o··a·ise·a·inedhise··t···ass·it
hs····iseasi·e·a·o·ationhadta·en··a·e·hi·e
he·asnt·oo·in·i·o··edso·e·o·e·ineandhesett
·ed·a··inhis·hai··a·eti·ted··to·a·dsthes·n
```

HNCMF now reads ·hai·. The candidate chair gives H → c and F → r.

Check every new letter assignment against earlier guesses and the one-to-one constraint.

Step 10: extend the partial solution

PARTIAL PLAINTEXT • · = UNDETERMINED

```
o·r·riend·ro··arise·a·inedhise··t···ass·it
hs·r·riseasi·e·a·orationhadta·en··ace·hi·e
he·asnt·oo·in·i·o·redso·e·ore·ineandhesett
·ed·ac·inhischair·aceti·ted··to·ardsthes·n
```

Visible fragment	Plausible completion	New mappings
o·r ·riend	our friend	I → u; Q → f
·ro· ·aris	from paris	V → m; E → p
·oo·in·	looking	X → l; S → k; T → g

Each completion is tested across the entire ciphertext before it is retained.

Step 11: complete the remaining letters

Apply the previous extensions first. The remaining unknowns occur inside strongly recognizable words.

Current fragment	Candidate word	New mapping
e·amined	examined	P → x
empt·	empty	B → y
·ith / ·hile / ·ine	with / while / wine	U → w
e·aporation	evaporation	K → v
·ack	back	G → b

These complete all 23 letter mappings observed in the message; check them at every occurrence.

The recovered message

Our friend from Paris examined his empty glass with surprise, as if evaporation had taken place while he wasn’t looking. I poured some more wine and he settled back in his chair, face tilted up towards the sun.

OBSERVED INPUT LETTERS

ABCDEFGHIKLMNOPRSTUVWXY

THEIR CIPHERTEXT IMAGES

CGHWZQTNMSXVRYEFDJIKUPB

J, Q, Z are absent: 3! = 6 full substitution tables still extend the 23 observed mappings.

Original reference example; spacing and punctuation restored by interpretation.

What the attack has taught us

We combined counts, pair statistics, repeated fragments, and plausible word boundaries.

Some decisions followed from the bijection constraint; others were statistical or contextual hypotheses.

A successful reconstruction explains all observed positions with one consistent substitution.

Exercise: identify one logical exclusion and one probabilistic guess in the walkthrough.

BREAK 01

Break / 10 minutes

We have seen how a fixed substitution preserves the language’s structure.

Next we change the substitution with position and ask which patterns remain.

Elapsed time: 60–70 minutes

03 / Vigenère cipher

Use several shift substitutions, selected by a repeating keyword.

Repeating-key polyalphabetic methods were described by Bellaso in 1553; the familiar name is associated with Vigenère.

Historical objective: weaken the direct link between letter identity and frequency.

Vigenère: spaces and key schedule

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = \mathbb{Z}_{26}^{m}$$

$$k = (k_0, \ldots, k_{m-1}), \qquad c_i = (p_i + k_{i \bmod m}) \bmod 26$$

$$p_i = (c_i - k_{i \bmod m}) \bmod 26$$

The keyword has m letters. Its effective repeating period can be smaller, as in ABCABC.

Encryption example: repeat the keyword

WRITE THE ALIGNMENT FIRST

```
Plaintext   CRYPTOGRAPHY
Keyword     LUCK
Key stream  LUCKLUCKLUCK
```

Position i	0	1	2	3	4	5
Plaintext value	2	17	24	15	19	14
Key value	11	20	2	10	11	20

Predict the first six ciphertext letters before continuing.

Encryption example: do the arithmetic

i	p_i	Key value	Sum mod 26	Ciphertext
0	2	11	13	N
1	17	20	11	L
2	24	2	0	A
3	15	10	25	Z
4	19	11	4	E
5	14	20	8	I

The wrap-around at i = 1 gives (17 + 20) mod 26 = 11, or L.

Complete the example and decrypt

COMPLETE ROWS

```
Plaintext   CRYPTOGRAPHY
Key stream  LUCKLUCKLUCK
Ciphertext  NLAZEIIBLJJI
```

$$p_1 = (11 - 20) \bmod 26 = 17 \quad (R)$$

Decryption subtracts the same aligned key value. Adding and subtracting cancel at each position.

R at positions 1 and 7 encrypts to L and B, because the aligned key letters differ.

Why does ordinary frequency analysis weaken?

A plaintext letter can be shifted in several different ways, depending on its position in the key schedule.

The aggregate histogram mixes these shifted distributions.

However, positions sharing the same key index still use one fixed shift.

The attack will isolate those positions instead of treating all letters as one population.

Cryptanalysis has two linked stages

Stage	Goal	Evidence / method
1	Find plausible periods	Repeated segments; index of coincidence
2	Recover the shifts	Frequency counts within aligned columns

Validate the combined result by decrypting the message. A plausible period alone does not identify the key.

Kasiski: why repeated segments can help

KNOWN ALIGNMENT EXAMPLE

```
Plaintext   THESUNANDTHEMANINTHEMOON
Key stream  KINGKINGKINGKINGKINGKING
Ciphertext  DPRYEVNTNBUKWIAOXBUKWWBT
```

THE begins at positions 9 and 17. Their distance 8 is divisible by the period 4, so both occurrences encrypt to BUK.

THE at position 0 has a different key alignment and encrypts differently.

Positions are zero-based. Same plaintext plus same key alignment implies the same ciphertext.

Kasiski: extract evidence from a ciphertext

Now use one longer ciphertext for the remaining attack: 1,045 normalized letters, with the key hidden.

ITS FIRST 126 LETTERS

```
ELQVVDXAFLZJESPCIFKETXUBTTWMZFZETWFESDMBQT
QCCWIRWAOMZUCLXFSEYWMGRYHQFZLCDSCWEOSRGIDM
YPXFSEHMFVNYSFVRCWKAOZPMBQSSBSGSEFHUPQQGFL
```

The complete ciphertext is printed in Appendix A1–A2. All positions and statistics below refer to that same text.

The repeated trigram FVR begins at positions 97, 147, 552, and 897.

Kasiski: measure the actual distances

Occurrence	Starting index	Distance from first
1	97	—
2	147	50
3	552	455
4	897	800

$$\gcd(50, 455, 800) = 5$$

The factors 1 and 5 are candidates. Chance repeats can mislead; check these candidates by a second method.

These distances are measured from the displayed dataset, not a separate hypothetical example.

Index of coincidence: the experiment

Let X be a string of N letters. Select an ordered pair of distinct positions uniformly at random.

IC is the probability that the two selected letters are equal.

$$N \geq 2, \qquad f_a = \#\{i : X_i = a\}, \qquad \sum_{a \in \mathbb{Z}_{26}} f_a = N.$$

This is sampling positions without replacement from the observed string.

Deriving the finite-sample IC formula

There are N(N − 1) ordered pairs of distinct positions.

For a letter a, exactly $f_a(f_a - 1)$ pairs have a in both positions. These events are disjoint across letters.

$$\mathrm{IC}(X) = \frac{\sum_{a=0}^{25} f_a(f_a - 1)}{N(N-1)}$$

$$= \sum_{a=0}^{25} \frac{f_a}{N} \frac{f_a - 1}{N - 1}.$$

A small IC calculation, fully worked

STRING

AABAAB

Letter	Count f_a	$f_a(f_a - 1)$
A	4	12
B	2	2
All others	0	0

$$\mathrm{IC}(\mathrm{AABAAB}) = \frac{12 + 2}{6 \cdot 5} = \frac{7}{15} \approx 0.4667.$$

Count matching pairs; do not divide by N^2 when positions must be distinct.

From sample counts to language probabilities

If independent letters have probabilities q_0, …, q_{25}, then the probability of equality is:

$$\Pr[X_i = X_j] = \sum_{a=0}^{25} q_a^2 \qquad (i \neq j).$$

Writing $\hat{q}_a = f_a/N$ gives an exact relation for the observed sample:

$$\mathrm{IC}(X) = \frac{N \sum_a \widehat{q}_a^2 - 1}{N - 1} \approx \sum_a \widehat{q}_a^2.$$

The approximation improves with N. Real language also has dependencies between positions.

Reference values and a useful invariant

Uniform independent letters: $\sum_{a=0}^{25} (1/26)^2 = 1/26 \approx 0.0385.$

English prose often has IC around 0.065, depending on the corpus and length.

$$\mathrm{IC}(\sigma(X)) = \mathrm{IC}(X) \quad \text{for any alphabet permutation } \sigma.$$

Reason: relabeling the alphabet merely rearranges the same counts in the sum.

A Caesar shift preserves IC exactly; the value is not a general measure of security.

Turn a candidate period into columns

$$X^{(\beta)} = (c_\beta, c_{\beta+m}, c_{\beta+2m}, \ldots), \qquad 0 \le \beta < m.$$

$$c_{\beta+jm} = (p_{\beta+jm} + k_\beta) \bmod 26 \quad \text{if } m \text{ is the period.}$$

Every symbol in a column has the same key shift, so each column should have language-like IC.

Wrong periods usually mix shifts; multiples of the true period can also separate them.

Make the columns explicit

FOR CANDIDATE PERIOD 5

$\beta = 0$: $C_0, C_5, C_{10}, C_{15}, \ldots$

$\beta = 1$: $C_1, C_6, C_{11}, C_{16}, \ldots$

$\beta = 2$: $C_2, C_7, C_{12}, C_{17}, \ldots$

$\beta = 3$: $C_3, C_8, C_{13}, C_{18}, \ldots$

$\beta = 4$: $C_4, C_9, C_{14}, C_{19}, \ldots$

For our 1,045-letter dataset, each of these five columns contains 209 letters.

Exercise: explain why a guessed period of 10 can also give language-like columns.

Compute IC for each candidate’s columns

Candidate m	Individual column IC values
1	0.0429
2	0.0416, 0.0441
3	0.0424, 0.0430, 0.0428
4	0.0426, 0.0452, 0.0402, 0.0449
5	0.0646, 0.0679, 0.0638, 0.0679, 0.0699

All five values at m = 5 are near the English reference range. This supports the Kasiski candidate 5.

Inspect individual columns; an average can conceal an inconsistent split.

One actual column: show the counts

FIRST 42 SYMBOLS OF COLUMN 0

EDZCTTZEQWOLYYLWGPHYCZSSPLWLOPSDLRGYCRNYPT

Symbol	P	Y	E	D	W	Total
Count	27	20	18	18	9	209

$$\sum_a f_a(f_a-1) = 2808, \quad \mathrm{IC}(X^{(0)}) = 0.0646.$$

The complete 26-symbol count table is in Appendix A3. The numerator uses every symbol, not just those shown here.

Recover the first shift: a candidate

P is the most frequent symbol in column 0, with 27 occurrences. If P is shifted E:

$$k_0 = (15 - 4) \bmod 26 = 11 \quad (L).$$

TRY SUBTRACTING 11

TSORIIOTFLDANNALVEWNROHHEALADEHSAGVNRGCNEI

A column is every fifth plaintext letter, so it will not read like a continuous sentence. Use its distribution to test the shift.

The most frequent-letter guess is a starting point; we next compare all 26 shifts.

Score a candidate using all letter counts

Let $O_a(k)$ count plaintext letter a after subtracting candidate shift k from the column.

$$E_a = N q_a, \qquad \chi^2(k) = \sum_{a=0}^{25} \frac{(O_a(k) - E_a)^2}{E_a}.$$

The q_a values form an illustrative English frequency model, normalized to sum to 1.

A smaller score means a closer fit to that model. This is a ranking heuristic, not proof that a key is correct.

The full model probabilities are included in Appendix A4.

Work out several score contributions

Plaintext a	Observed $O_a(11)$	Expected Nq_a	Contribution
E	27	26.547	0.008
N	20	14.106	2.463
A	18	17.069	0.051
Z	0	0.155	0.155

$$\chi^2(11) = \sum_{a=0}^{25} \text{contribution}_a = 22.297.$$

Sum all 26 terms. The four rows above are selected contributions, not the complete sum.

Compare the candidate shifts

Shift k	Key letter	Full χ^2 score
11	L	22.297
24	Y	526.998
22	W	699.130
17	R	839.822

Shift 11 gives the smallest score among all 26 candidates for this column.

Repeat the same calculation independently for the other four columns.

This avoids enumerating the entire product of 26^5 possible five-letter keys.

Recover all five key letters

Column β	Best shift	Key letter	Full χ^2 score
0	11	L	22.297
1	4	E	29.388
2	12	M	28.459
3	14	O	23.656
4	13	N	23.953

The resulting key hypothesis is LEMON. We must now test it on the interleaved complete message.

Validate by decrypting the full message

FIRST 42 LETTERS

```
Ciphertext   ELQVVDXAFLZJESPCIFKETXUBTTWMZFZETWFESDMBQT
Plaintext    THEHISTORYOFSECRETWRITINGISALSOAHISTORYOFP
```

Restoring word boundaries gives “The history of secret writing is also a history…”

The complete recovered prose is printed in Appendix A5–A6. It uses one key consistently across the entire message.

Re-encryption checks consistency; coherent full-message decryption supplies additional evidence.

What did each stage establish?

Evidence	Conclusion supported
Repeated FVR; gcd of distances is 5	Candidate periods include 1 and 5.
Individual IC values near English at m = 5	Period 5 is statistically plausible.
Column-wise frequency scores	A specific key hypothesis: LEMON.
Coherent complete decryption	The combined hypothesis explains this dataset.

Short texts, unusual language, and accidental repeats can make every statistical stage less decisive.

BREAK 02

Break / 10 minutes

The attack used arithmetic, probability, and language evidence together.

Next we replace single-letter substitution with pairs, then rearrange positions.

Elapsed time: 120–130 minutes

04 / Playfair cipher

A compact construction for encrypting letter pairs by hand.

Invented by Charles Wheatstone in 1854 and promoted by Lyon Playfair.

Historical attraction: pair substitution without carrying a huge substitution table.

Why substitute more than one letter?

Single-letter substitution leaves each letter’s frequency directly visible under another label.

Playfair works with **digrams**: pairs of distinct letters prepared from the message.

Codebooks could replace entire words or instructions, but large books were harder to distribute and protect.

This is polygraphic substitution; Vigenère instead changes a single-letter mapping with position.

Build the square: normalize the keyword

KEYWORD

PLAYFAIR EXAMPLE

KEEP FIRST OCCURRENCES ONLY

P L A Y F I R E X M

Merge J into I, remove spaces, and scan left to right, skipping any letter already entered.

These ten distinct letters occupy the first two rows of the 5 × 5 square.

Build the square: fill the remaining cells

First enter the keyword letters.

P	L	A	Y	F
I	R	E	X	M

Then append unused letters in order.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Unused letters: B C D G H K N O Q S T U V W Z. Both parties build the same square.

Prepare the message, one pair at a time

Remaining text	Action	Pair emitted
BALLOON	B and A differ; consume both.	BA
LLOON	Repeated L: consume first L; insert X.	LX
LOON	L and O differ; consume both.	LO
ON	O and N differ; consume both.	ON

Thus BALLOON becomes BA LX LO ON. Do not consume both repeated L letters when inserting the filler.

Preparation conventions and the gold example

If one letter remains, append X. For a repeated or final X, use Q as the alternate filler.

EXAMPLES

`X → XQ` `XX → XQ XQ`

HIDE THE GOLD IN THE TREE STUMP

`HI DE TH EG OL DI NT HE TR EX ES TU MP`

Only letters within the next proposed pair trigger the repeated-letter rule.

Preparation can lose information; we keep it separate from the reversible pair transformation.

Rule 1: letters in the same row

1. Locate P and L in row 1.
2. Move each one cell to the right.
3. Preserve their order.

RESULT

```
PL → LA
Wrap-around: FP → PL
```

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Grid shading marks the input pair. Decryption moves left instead of right.

Rule 2: letters in the same column

1. Locate P and I in column 1.
2. Move each one cell down.
3. Wrap from the bottom to the top.

RESULT

```
PI → IB
Wrap-around: TP → PI
```

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Decryption moves up; both row and column rules wrap around the square.

Rule 3: opposite rectangle corners

H is at (row 3, column 5).

I is at (row 2, column 1).

Keep each input row; exchange the columns.

RESULT

```
H → B at (3,1)
I → M at (2,5)
HI → BM
```

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

The first output stays in the first input’s row; pair order matters.

Encrypt the prepared message in stages

Input pair	Relation in the square	Output
HI	Rectangle	BM
DE	Same column	OD
TH	Rectangle	ZB
EG	Rectangle	XD

Work through these first four pairs before applying the same rules to the rest.

Complete the encryption and reverse it

PREPARED PLAINTEXT

HI DE TH EG OL DI NT HE TR EX ES TU MP

CIPHERTEXT

BM OD ZB XD NA BE KU DM UI XM MO UV IF

For decryption: move left in a row, up in a column, or apply the same rectangle rule.

Check BM → HI and OD → DE using the square.

Exercise: decrypt with the same square

Pair exercise • 3 minutes

CIPHERTEXT

```
RO NC FL WI
```

Decrypt each pair and name the rule used.

Keep any filler until you have recovered all the prepared letters.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

The complete calculation follows on the next slide.

Solution: show every pair and rule

Ciphertext	Inverse rule	Prepared plaintext
RO	Rectangle	EN
NC	Same column: move up	CR
FL	Same row: move left	YP
WI	Rectangle	TX

RESULT

```
EN CR YP TX → ENCRYPTX
```

The intended reading is ENCRYPT, with a final X supplied during preparation.

Why exact spelling may not be recoverable

TWO DIFFERENT ORIGINALS

```
AXA → AX AX
AA  → AX AX
```

Both originals produce the same prepared text and therefore the same ciphertext.

Merging I and J also loses information. Do not automatically remove every X from a decryption.

The pair transformation is invertible on prepared pairs; preprocessing need not be invertible.

Historical usefulness and cryptanalysis

Playfair offered a practical balance between manual operation and more complex substitution; it saw military field use.

Pair and longer language patterns still survive. A computer can decrypt with candidate squares and score the results.

HILL-CLIMBING OUTLINE

```
square → decrypt → language score
modify square → keep improvement → repeat
```

Random restarts help with local optima. Success depends on the message, language model, and search effort.

Understand the historical design goal before judging it by present-day requirements.

05 / Transposition cipher

Keep letter identities unchanged and permute their positions within each block.

Substitution changes symbols; transposition changes where they occur.

Permutation cipher: formal definition

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{m}, \qquad \mathcal{K} = S_m$$

$$E_\pi(p_1, \ldots, p_m) = (p_{\pi(1)}, \ldots, p_{\pi(m)})$$

$$D_\pi(c_1, \ldots, c_m) = (c_{\pi^{-1}(1)}, \ldots, c_{\pi^{-1}(m)})$$

Apply this operation block by block. Positions here are explicitly numbered 1 through m.

For incomplete blocks, pad with X and retain the original message length separately.

Read the permutation in the right direction

Output position i	1	2	3	4	5	6
Input position π(i)	3	1	4	6	2	5

The i-th ciphertext letter comes from input position π(i).

FIRST THREE OUTPUTS

$c_1 = p_3 \qquad c_2 = p_1 \qquad c_3 = p_4$

Some sources define the reverse direction. This table fixes the convention used throughout our example.

Encrypt one block, step by step

Output position	1	2	3	4	5	6
Read input position	3	1	4	6	2	5
Input letter from ATTACK	T	A	A	K	T	C

TWO-BLOCK MESSAGE

```
ATTACK ATDAWN
TAAKTC DAANTW
```

Every letter appears exactly as often after rearrangement as before it.

Construct the inverse permutation

Original input j	1	2	3	4	5	6
Output position containing j	2	5	1	3	6	4

READ THE INVERSE

$\pi^{-1} = [2, 5, 1, 3, 6, 4]$

TAAKTC → ATTACK

$$c_{\pi^{-1}(j)} = p_{\pi(\pi^{-1}(j))} = p_j.$$

The equality explains why the inverse table undoes encryption.

Exercise: encrypt, then explain an invariant

Individual calculation • 2 minutes

PLAINTEXT AND KEY

DEFEND

$\pi = [3, 1, 4, 6, 2, 5]$

Encrypt the block and recover it using the inverse permutation.

How many keys are possible at block size 6? Which letter statistics are unchanged?

Use the declared output-to-input convention, not a memorized alternative.

Solution and security discussion

SOLUTION

DEFEND → FDEDEN → DEFEND

$$|S_6| = 6! = 720, \qquad f_C(a) = f_P(a) \text{ for every letter } a.$$

For this block size, exhaustive search is small. Candidate rearrangements can also be ranked using language patterns.

With larger blocks there are more keys, but unchanged counts remain an important structural property.

SYNTHESIS

Compare the five constructions

Cipher	Mathematical key	Information exploited in an attack
Shift	$k \in \mathbb{Z}_{26}$	Small key space
Substitution	$\sigma \in S_{26}$	Letter and word-pattern statistics
Vigenère	$k \in \mathbb{Z}_{26}^{m}$	Periodicity and column statistics
Playfair	A keyed 5 × 5 square	Pair and longer language statistics
Transposition	$\pi \in S_m$	Preserved counts and positional patterns

Each construction illustrates a different relationship between a transformation and its invariants.

SYNTHESIS

From historical alphabets to digital messages

Our hand-cipher examples first removed spaces and punctuation. That preprocessing prevented exact recovery of the original formatting.

Digital encryption operates on encoded bytes or bits and can recover the complete original representation.

The next lectures develop stronger constructions and additional security goals. These five ciphers establish the language and methods for that study.

Historical usefulness and present-day suitability are different questions.

SYNTHESIS

Reflection questions

Closing discussion • 3 minutes

1. Why is the inverse map necessary in a substitution cipher?

2. Which step of the frequency attack was a logical exclusion, and which was a statistical guess?

3. Derive the denominator of IC and explain why a shift preserves IC.

4. Explain why candidate periods 5 and 10 can both look plausible.

Use definitions, calculations, and examples to support your answers.

REFERENCES

Reading and sources

Core reading: Stinson & Paterson, *Cryptography: Theory and Practice*, 4th edition, Chapter 1. Consult the classical-cipher sections in your edition.

Lecture reference: Wu Hongjun, supplied *Lecture02_Classical_Ciphers.pdf*. The 168-letter substitution example and standard Playfair example are retained.

History: Kahn, *The Codebreakers* (revised edition, 1996); Kerckhoffs (1883); Kasiski (1863).

Appendix: complete Vigenère ciphertext, column counts, language model, and recovered prose. The model is illustrative and corpus dependent.

End of the scheduled lecture. The following data pages support the worked calculations.

APPENDIX

A1 / Complete Vigenère ciphertext

POSITIONS 0–527 • ZERO-BASED

```
ELQVVDXAFLZJESPCIFKETXUBTTWMZFZETWFESDMBQTQCCWIR
WAOMZUCLXFSEYWMGRYHQFZLCDSCWEOSRGIDMYPXFSEHMFVNY
SFVRCWKAOZPMBQSSBSGSEFHUPQQGFLKQKVWPDSZLMZVVOHQB
LPXFVRSENWGDSROYLRSINRIEIEGMHSVYWGFCCMEWARAMMFNS
YABYPQHGPVEGGTPXCPNYDCSEIZFRAIMHROAAFQDPQOIPVQDR
LXQRFSEBSFLRPTNXMXWNCTTFNDIEDEZZURRNPGSFLGMFRQYX
FRLHQFQZIEBBEXDSNEEZMFTRSZRNPGSNDTDCBQMZGGPEPHUP
VQOQPVYOXPWMUHPWEQUPGWGVEWOCADICIRYGQGNYHOVNYKQG
GSISIRDWIVRYXTSRGMPSANIPWFLKDSRDGAACFXQFFXEWSGSM
EDEZGQGFQEEHRCFGHQZRAHEPQAJRELQBRPHRCELGXSNCQARR
WSRHUPTDCOWIYKUPRMGUZVFYRJMEFRAIMHROEOFBDWMZBYKY
```

Join both pages without spaces or line breaks. All Kasiski positions and IC values refer to this 1,045-letter string.

APPENDIX

A2 / Complete Vigenère ciphertext

POSITIONS 528–1044 • ZERO-BASED

```
SFDESSGSIEOZPTAGVEMABFTRFVRVIKFREYDBNREUBNYHMUNT
REDYTXFWARXTSZPWEOTPMZHBNSXIZYWOOACIHSNWEEWZAPQD
NEXQFATREWQPIMQUNSXIZYIZCHRLFSKEKUJRDXTSNYEXMFEE
NSGEIDQULROSBQHUGGTRSIVDLUBTLYESSFPBOGEIDBSCSYOP
ZMZQVOIZQRELQZRDWABVDXTOGLPMFTPRGAOPVATCZWEWOWIW
SLDMECAWCABRAEDHBQWQQHCMFMJPQGGGLPECNDOIVNEMZTBC
QMHVZRDSZLMZGITWUPYPEZRJSEFOAZFESEGIDQNYPQOEYJDC
ZTXSCBOIZUVYIQFVYKEHNCXEPLDXMHVYKFVRLWEIZAXUCADG
TSPVMZURIEYDYPWMBQWSAYVYKRCEHEKGGSEFOFJWFSZNEZTN
TPMGRNYDSQPWUUAXYEHFFVHWIPGMFRQYXDHMPUQFEYPMELXT
SEELMBQPTQBQZRWSRAMZUGSIYSGSSPIAVRAKA
```

Join both pages without spaces or line breaks. All Kasiski positions and IC values refer to this 1,045-letter string.

APPENDIX

A3 / All counts for Vigenère column 0

Letter	Count	Letter	Count	Letter	Count	Letter	Count
A	6	B	0	C	11	D	18
E	18	F	3	G	4	H	2
I	1	J	2	K	0	L	18
M	1	N	9	O	6	P	27
Q	7	R	5	S	11	T	12
U	0	V	3	W	9	X	3
Y	20	Z	13				

These 26 counts sum to 209. Use all of them when computing IC or scoring candidate shifts.

APPENDIX

A4 / Illustrative English frequency model

Letter	q_a	Letter	q_a	Letter	q_a	Letter	q_a
A	0.08167	B	0.01492	C	0.02782	D	0.04253
E	0.12702	F	0.02228	G	0.02015	H	0.06094
I	0.06966	J	0.00153	K	0.00772	L	0.04025
M	0.02406	N	0.06749	O	0.07507	P	0.01929
Q	0.00095	R	0.05987	S	0.06327	T	0.09056
U	0.02758	V	0.00978	W	0.02360	X	0.00150
Y	0.01974	Z	0.00074				

Rounded for display. Calculations use normalized unrounded values; corpus frequencies vary.

APPENDIX

A5 / Recovered Vigenère prose, part 1

The history of secret writing is also a history of people finding patterns. A sender may replace every letter with another symbol and hope that the message will remain hidden. Yet the habits of a language survive in surprising ways. Common letters still occur often, repeated words leave repeated shapes, and familiar phrases provide clues. A careful reader does not treat any single clue as proof. Instead the reader makes a guess, checks its consequences, and changes the guess when the evidence disagrees. Computers make this process faster but do not remove the need for a clear model of the problem.

Word boundaries and punctuation are editorial reconstructions, not encoded by the cipher.

APPENDIX

A6 / Recovered Vigenère prose, part 2

When a short key is repeated across a long message, the same positions in the key return again and again. Splitting the message into columns can reveal a simple pattern inside each column. Enough text gives the analyst a better chance of distinguishing a useful pattern from a coincidence. The lesson is that a large number of possible keys is only one part of security. We must also ask what information remains visible and what an observer can learn from it. Good engineering starts by stating the assumptions, checking examples, and looking for ways that a system can fail. A secure design must survive careful public study rather than depend on keeping the method unknown.